

Structural Models in Household Finance

Sylvain Catherine

What is household finance?

The financial decisions of households — and of the markets that serve them

The map of the field (Gomes et al., 2021)

- **Assets:** saving motives, participation, asset allocation, trading, retirement saving and decumulation
- **Liabilities:** mortgage choice, refinancing, default; credit cards and payday lending
- **Social environment:** peers, culture, heredity, intra-household bargaining
- **Literacy, cognition, and advice**

A relatively young field

- Named by [Campbell \(2006\)](#); NBER working group 2009, CEPR network 2015, JEL code G50 in the early 2020s
- **Why so late? Two hard prerequisites:**
 - *A normative benchmark:* life-cycle models with constraints, uninsurable income, flexible labor supply, endogenous retirement
⇒ large state space
 - *Positive evidence:* from small surveys to administrative panels, plus credible causal inference

How household finance connects to other fields

- **Macroeconomics** — *households are the transmission channel*
 - MPC heterogeneity and the wealthy hand-to-mouth; HANK
 - Monetary policy as redistribution between borrowers and lenders; refinancing inertia weakens it where it is most needed
- **Asset pricing** — *households ultimately bear aggregate risk*
 - Limited participation: the marginal investor is not the average household → equity premium
 - Background risk — labor income, housing, private business — sets how much risk households can bear
- **Public economics** — *policy shapes the balance sheet*
 - Consumption responses identify marginal utility → welfare analysis of UI, health, disability insurance
 - Attention to tax incentives varies across households
- **Inequality and the supply side** — returns and borrowing costs differ sharply across the wealth distribution; and firms often exploit household mistakes rather than correct them → IO and consumer protection

Plan for today: one question, two methods

Question: how does labor income risk shape household portfolios?

1. **Part 1 — A structural model** ([Catherine, 2022](#))

Life-cycle portfolio choice when the skewness of idiosyncratic income shocks is cyclical.
What the model is, how it is estimated, and what it delivers that no regression could.

2. **Part 2 — Structure informing reduced form** ([Catherine et al., 2024](#))

The model tells you *what* to measure, *how* to scale it, and *what a placebo looks like*.
Swedish registry data then test the mechanism directly.

PART 1

A STRUCTURAL MODEL

How much should households invest in stocks?

Countercyclical Labor Income Risk and Portfolio Choices over the Life Cycle

Sylvain Catherine

Review of Financial Studies, 2022, 35(9), 4016–4054

The oldest literature in household finance — and its problem

- **The oldest strand of the field.**
 - Analytical lifetime portfolio problem: [Samuelson \(1969\)](#) and [Merton \(1969\)](#)
 - Computational models with background risk: [Viceira \(2001\)](#), [Cocco et al. \(2005\)](#)
- **And the most influential normatively.** The declining equity glide path of target-date funds — now the default investment of most US 401(k) participants — is a direct descendant of these models
- **But as a *positive* model it does not merely fit poorly — it predicts the opposite:**
 - Model: the equity share *falls* with age.
 - Data: participation and the conditional equity share *rise* until retirement
 - **A benchmark with a negative R^2 against the very moments it was built to explain**
- **Two readings, opposite implications:**
 - (a) Households make large mistakes → the normative advice stands, and should be nudged
 - (b) The model omits background risk households perceive and we do not measure → the advice is wrong

Theory – Optimal equity share over the life cycle

- Optimal equity share without labor income

$$\pi = \frac{\mu - r}{\gamma \sigma^2}$$

→ π does not depend on age (Merton, 1969)

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$$\frac{\pi W}{W + H} = \frac{\mu - r}{\gamma \sigma^2} \Leftrightarrow \pi = \frac{\mu - r}{\gamma \sigma^2} + \frac{\mu - r}{\gamma \sigma^2} \frac{H}{W}$$

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- Risky human capital

$$\frac{\pi W + \beta_H H}{W + H} = \frac{\mu - r}{\gamma \sigma^2} \Leftrightarrow \pi = \frac{\mu - r}{\gamma \sigma^2} + \left(\frac{\mu - r}{\gamma \sigma^2} - \beta_H \right) \frac{H}{W}$$

→ age profile depends on γ and the market beta of human capital β_H

A very hard measurement problem: β_H

- How do we measure the market beta of human capital?
 - Theory: “regress human capital returns” $r_{H,it}$ on stock returns to estimate:

$$\beta_H = \frac{\text{Cov}(r_H, r_s)}{\sigma_{r_s}^2}$$

- What is r_H ? Just like for stock returns, it should be

$$r_H = \frac{H_{it+1} - H_{it} + Y_{it}}{H_{it}}$$

Empirical challenge: the only thing we observe is Y_{it}

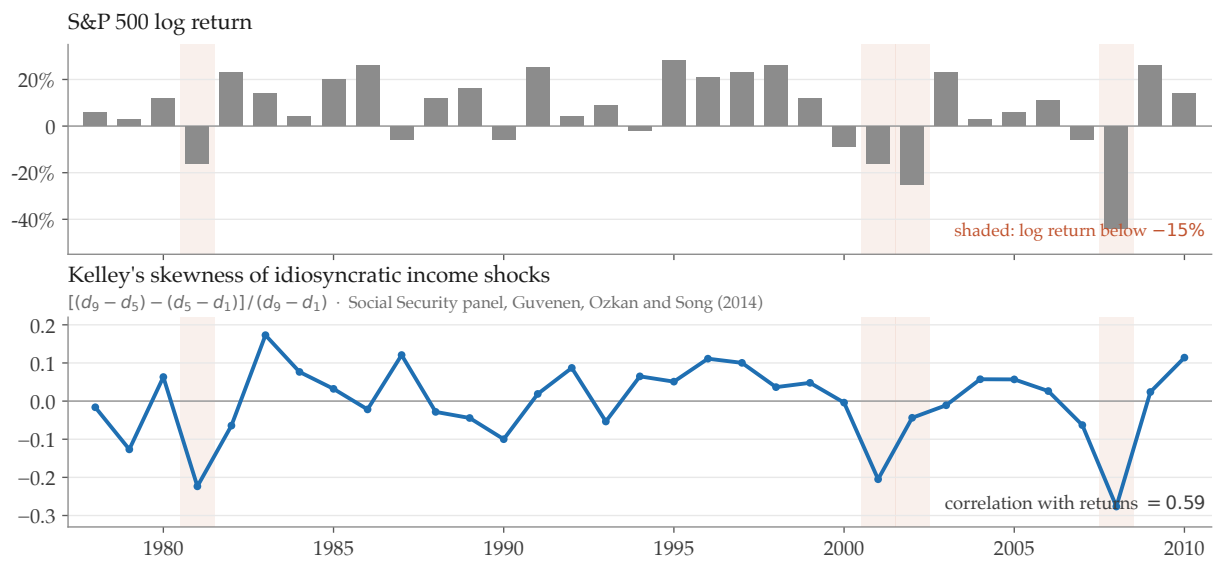
- **Solution:** go structural, parametric assumption about the stochastic properties of Y_{it} .

Seminal papers by [Viceira \(2001\)](#) and [Cocco et al. \(2005\)](#):

- Calibrate correlation between shocks to Y_{it} and market returns to match data $\Rightarrow$ close to zero correlation ($\beta_H = 0$)
 - Most working-age households should invest most of their wealth in stocks, even with very high relative risk aversion ($\gamma = 10$)

Correlation isn't everything!

Income risk skewness and stock returns



Mean shocks are barely correlated with returns — but the *left tail* of the income distribution opens up exactly in the years the market crashes.

The model (Catherine, 2022)

- **Preferences.** CRRA over a bundle of goods and housing:

$$V_{t_0} = \mathbb{E} \sum_{t=t_0}^T \beta^{t-t_0} \left(\prod_{k=t_0}^{t-1} (1 - m_k) \right) \frac{(C_{it}^{1-\nu} H_{it}^\nu)^{1-\gamma}}{1-\gamma}$$

Choices: consumption, rent vs. own, and the equity share π of net worth.

- **Income.** Persistent component of earnings $z_{it} = \rho_z z_{it-1} + \zeta_{it}$ and innovations drawn from a *normal mixture*:

$$\zeta_{it} \sim \mathcal{N}(\mu_{z,t}^-, \sigma_z^{-2}) \text{ w.p. } p_z, \quad \zeta_{it} \sim \mathcal{N}(\mu_{z,t}^+, \sigma_z^{+2}) \text{ w.p. } 1 - p_z$$

- **The one innovation:** the mean of the bad component moves with the aggregate wage index,

$$\mu_{z,t}^- = \mu_z^- + \lambda_{zl} (l_{1,t} - l_{1,t-1}), \quad \lambda_{zl} > 0$$

→ left-tail career risk spikes exactly when the economy — and the stock market — is down.

- **Realism elsewhere:** Social Security benefits based on lifetime earnings, a means-tested safety net, stock market crashes, homeownership with down payment and transaction costs, a fixed participation cost.

Step 1: estimate the income process outside the portfolio problem

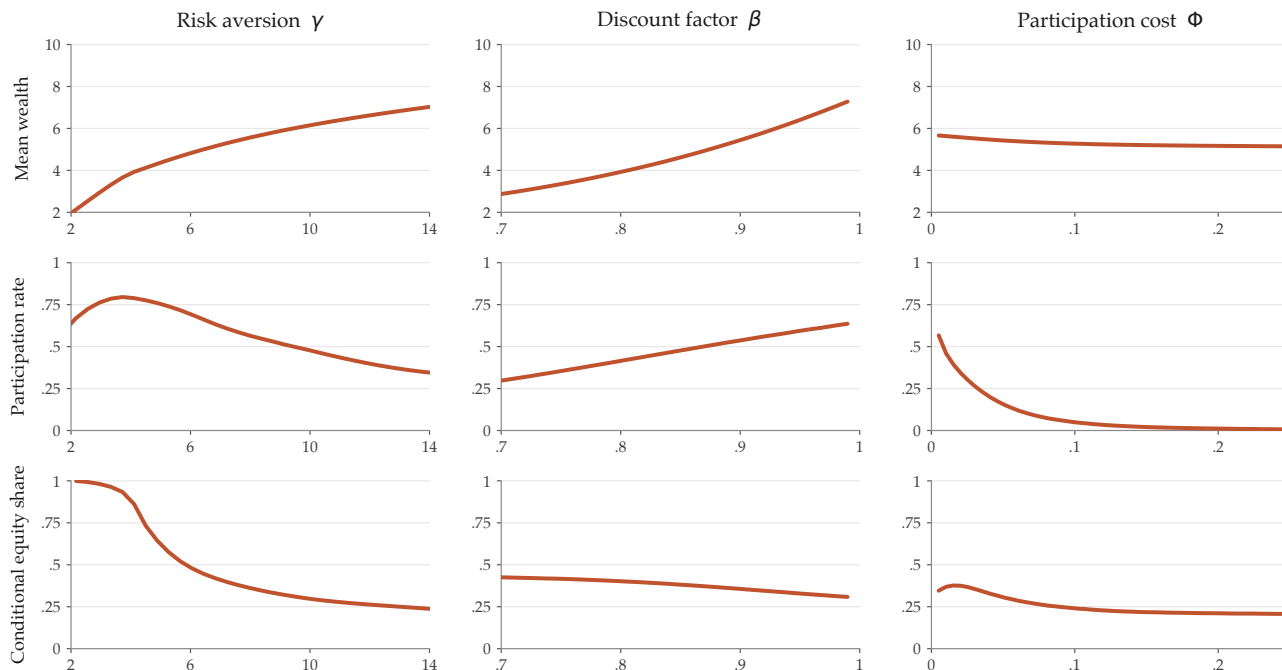
- **Targets.** Cross-sectional mean, variance, and skewness of idiosyncratic earnings growth, *for each year* 1978–2010, from Social Security administrative panel data ([Guisen et al., 2014](#))
- The same moments at 1-, 3-, and 5-year horizons disentangle transitory from persistent shocks and identify the persistence ρ_z
 - 155 empirical moments; 11 macro and 8 idiosyncratic parameters
 - Finding: **tail shocks are persistent** → they are the kind of risk that matters for portfolios
- **Note what is *not* used here:** no portfolio data, no wealth data

Design principle: estimate what you can from data the model is not being asked to explain. Parameters estimated in step 1 are no longer degrees of freedom in step 2 to fit the moments at the heart of your research question. Also practical since the income process is far less costly to simulate.

Step 2: estimate preferences by SMM

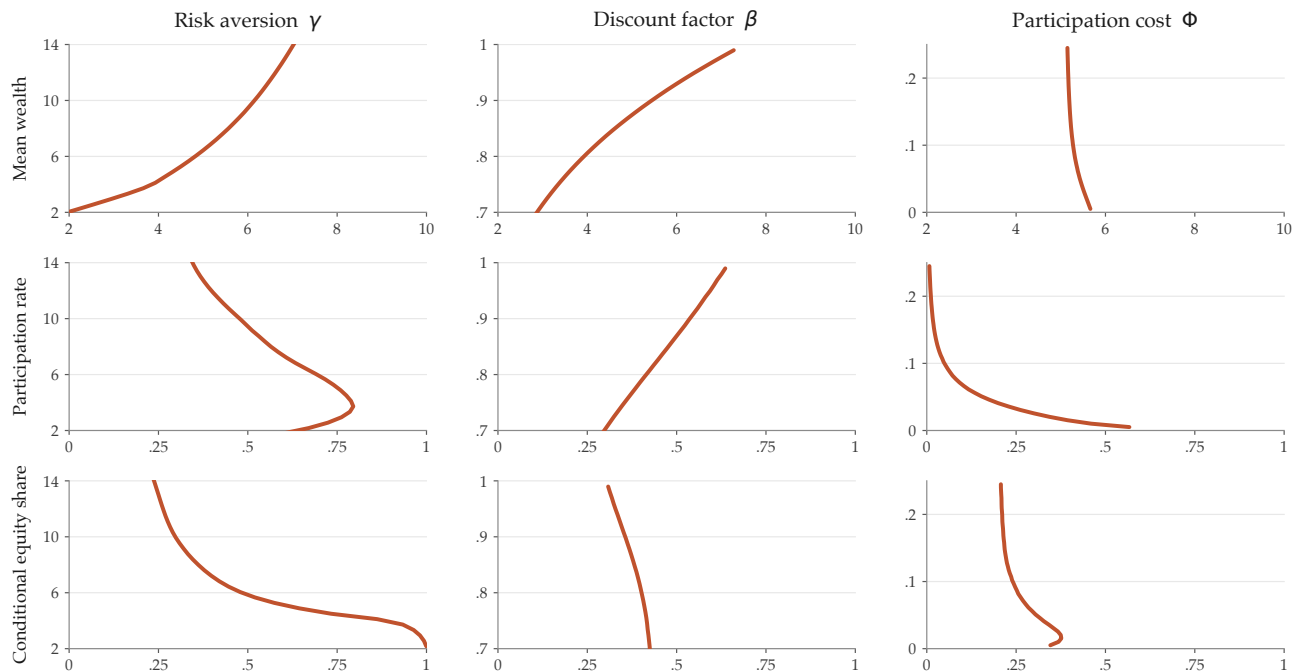
- **Parameters:** risk aversion γ , discount factor β , fixed yearly participation cost c_π
- **Moments:** SCF life-cycle profiles of wealth, participation, and the conditional (or unconditional) equity share, for age groups $[23, 25]$ to $[62, 64] \Rightarrow 56$ moments
- **Identification — one line per parameter:**
 - $\beta \leftarrow$ the level and slope of wealth accumulation over the life cycle
 - $\gamma \leftarrow$ the level of the equity share among participants
 - $c_\pi \leftarrow$ the participation rate, especially among young, low-wealth households

Step 2: How do the moments move with the parameters?



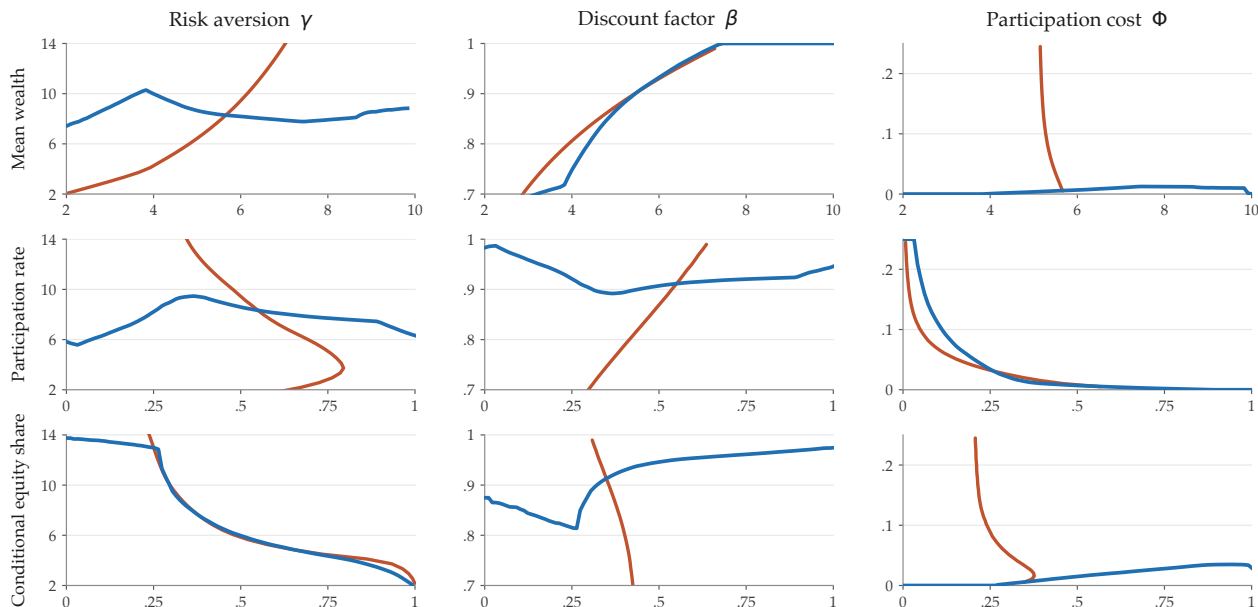
each panel: value of the parameter on the x-axis, value of the moment on the y-axis

Step 2: Does inverting comparative statics convey identification intuition?



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Step 2: Inverting the SMM itself to check identification intuition

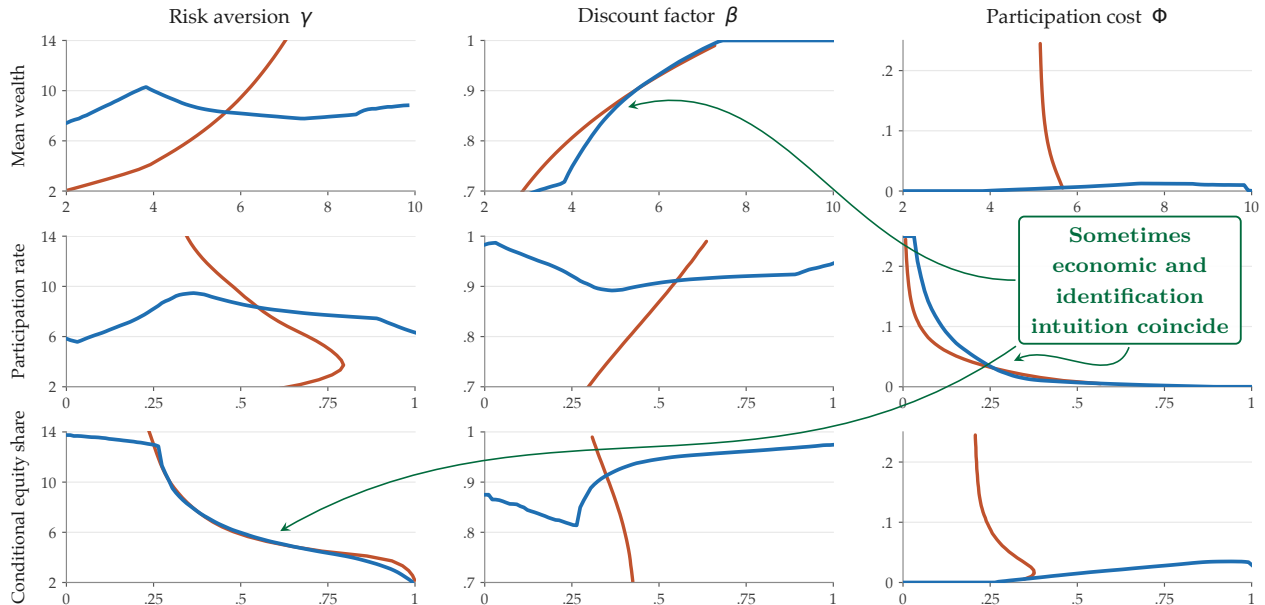


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— Estimate response: $\hat{\theta}$ when the targeted moment moves

— Inverted comparative statics: $f(\theta)$

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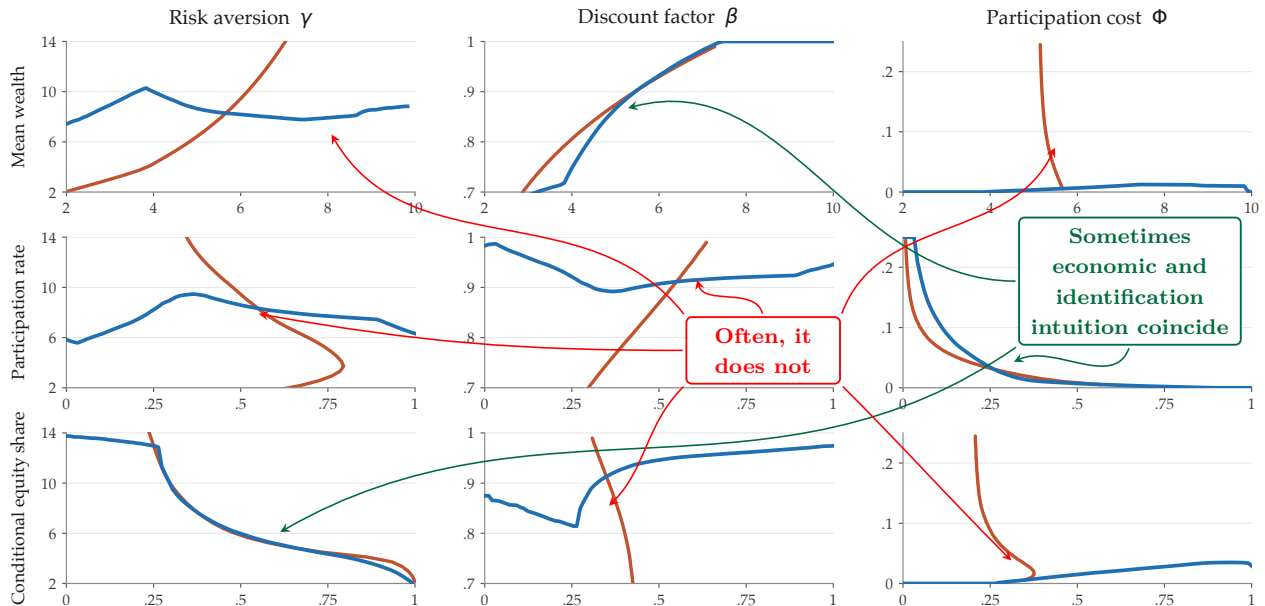


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How do you invert an SMM?

- **The obstacle.**
 - Any variation in moments is a *brand-new* estimation, each solving the model thousands of times
 - The three figures we just saw need hundreds of separate re-estimations. Often prohibitively expensive
- **The trick: lightspeed surrogate SMM**
(Catherine, Ebrahimian, Fereydounian, Sraer and Thesmar, 2026)
 1. Draw many parameter vectors and solve the model *once* for each → a cloud of (parameters, moments) pairs
 2. Fit a flexible approximation (e.g. neural nets) of the parameters → moments on that cloud
 3. Re-estimating using the approximation is now evaluating a function: essentially free
- **What it buys.** Identification curves, sample splits, moment selection, misspecification checks — routine instead of infeasible. The paper gives the convergence conditions, and validates the approximation out of sample

All the identification figures in this part are taken from this working paper.

Back to the paper: what cyclical skewness buys

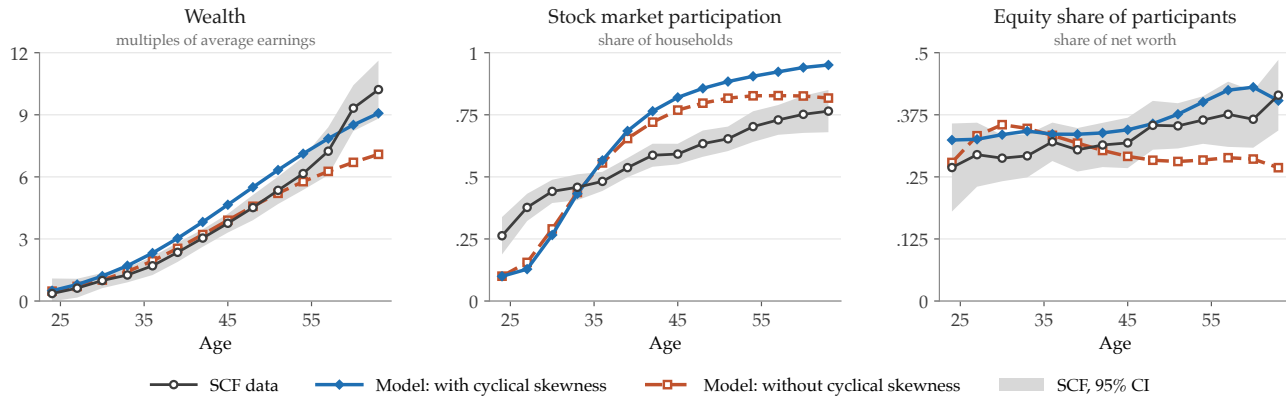
	With cyclical skew.		Without	
	No cost	With cost	No cost	With cost
Relative risk aversion γ	7.14 (0.042)	6.07 (0.042)	11.95 (0.120)	8.01 (0.068)
Discount factor β	0.911 (0.008)	0.956 (0.005)	0.666 (0.022)	0.882 (0.010)
Participation cost	—	\$250 (<0.001)	—	\$400 (<0.001)

Standard errors in parentheses. Participation cost is estimated as a share of the average wage.

- Without cyclical skewness, human capital is bond-like
 - Agents want to buy lots of stocks $\Rightarrow$ SMM returns a very high risk aversion to discourage them
 - But high risk aversion means lots of precautionary savings $\Rightarrow$ SMM returns very low discount factor to keep it close to empirical moment
- Not a new problem: [Fagereng et al. \(2018\)](#)-style findings in Norwegian data ($\gamma = 11$, $\beta = 0.77$)

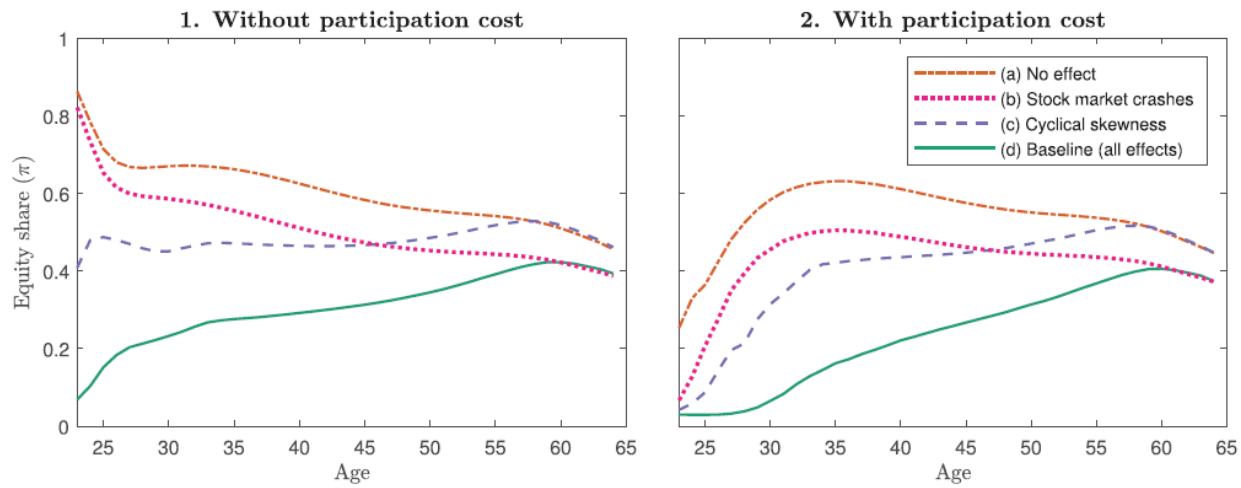
A single feature of the income process moves the implied preferences from implausible to reasonable.

Model fit: targeted life-cycle moments



- Both specifications fit **wealth** and the **extensive margin** about equally well — targeting those alone would not separate them
- The **conditional equity share** is where they part company: without cyclical skewness the model produces a *declining* profile, the data and the estimated model both rise

Decomposing the mechanisms



PART 2

STRUCTURE INFORMS REDUCED FORM

Do households actually hedge cyclical skewness?

Countercyclical Income Risk and Portfolio Choices: Evidence from Sweden

Sylvain Catherine Paolo Sodini Yapei Zhang

Journal of Finance, 2024, 79(3), 1755–1788

The empirical literature — and its two problems

- **The core prediction of portfolio theory is simple:** households whose human capital is more exposed to stock market risk should hold less equity
- **Yet the evidence is mixed at best** (Vissing-Jorgensen, 2002; Massa and Simonov, 2006; Calvet and Sodini, 2014; Fagereng et al., 2018). Two reasons:
 - **Measurement.** These papers regress equity shares on the *variance* or the *covariance* of labor income. But theory gives variance an *ambiguous* sign, and the covariance is ≈ 0 in the data
 - **Endogeneity.** Risk-tolerant workers take riskier jobs *and* buy more stocks
- **And there is no instrument.** How would you instrument for the cyclical risk of a worker's tail income risk? No policy experiment, no plausible shifter, no discontinuity
- **Why the structural model helps:**
 - It says *which* moment of the income distribution should matter
 - It delivers *comparative statics*: how the coefficient must vary with observables

A hypothesis that had never been tested directly

- **An old idea.** That countercyclical idiosyncratic risk explains portfolio and asset pricing puzzles goes back to [Mankiw \(1986\)](#)
- **A large quantitative literature relies on it.**
Portfolio choice: [Storesletten et al. \(2007\)](#), [Lynch and Tan \(2011\)](#), [Catherine \(2022\)](#)
Asset pricing: [Constantinides and Duffie \(1996\)](#), [Schmidt \(2016\)](#), [Constantinides and Ghosh \(2017\)](#)
- **But no direct evidence.** No one had shown that households facing more left-tail income risk in downturns actually hold fewer stocks

What the structure says to measure

Start from the portfolio rule:

$$\pi W = \frac{\mu - r}{\gamma \sigma_s^2} W + \frac{\mu - r}{\gamma \sigma_s^2} H - \beta_H H, \quad \beta_H = \frac{\text{Cov}(r_H, r_s)}{\sigma_s^2}$$

The third term: how many dollars of “stock” are already embedded in human capital.

β_H loads on three distinct comoments, not one:

- **Covariance risk:** $\text{Cov}(\text{mean of income shocks}, r_s)$ — close to zero in the data (Cocco et al., 2005)
- **Countercyclical variance:** $\text{Cov}(\text{variance}, r_s)$ — little support in US administrative data (Guisen et al., 2014)
- **Cyclical skewness:** $\text{Cov}(\text{third moment}, r_s)$ — the salient feature of the data (Guisen et al., 2014)

...and what the coefficient should look like

Two predictions, both read straight off the portfolio rule:

- **Prediction 1.** Higher covariance, countercyclical variance, or cyclical skewness risk $\Rightarrow$ lower equity share
- **Prediction 2.** The effect scales with the human capital share of total wealth, $H/(W + H)$. Therefore:
 - it *shrinks* as workers approach retirement
 - it is *zero* for workers whose $H/W \rightarrow 0$ — a theoretical placebo group
 - the relationship should pass through the origin in $H/(W + H)$

This is the real payoff. An omitted variable — risk preference, literacy, upbringing — could produce Prediction 1. To produce Prediction 2 it would have to be moderated by the human capital share in exactly the way theory specifies.

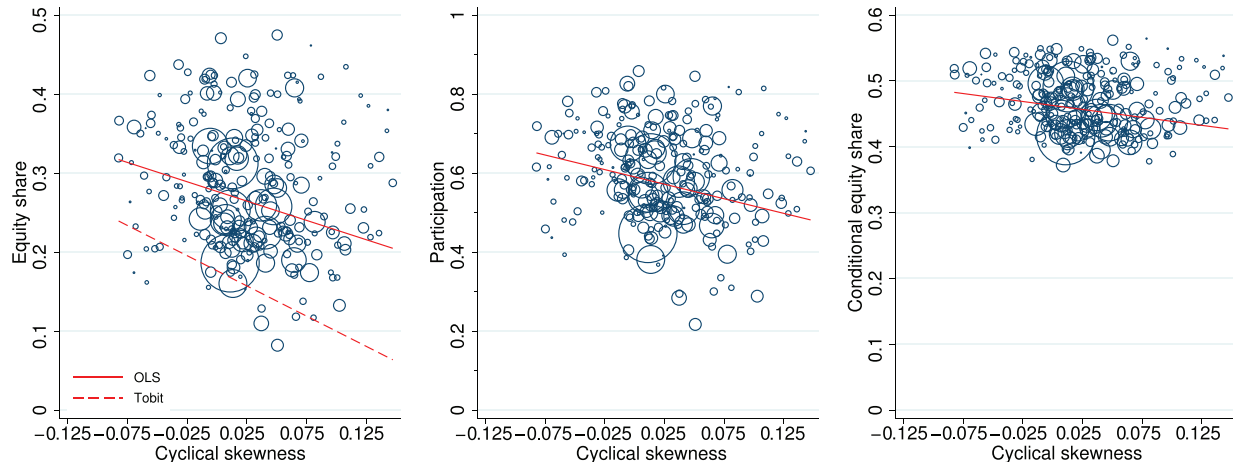
Design: Catherine et al. (2024)

- **Data.** Swedish administrative panel on income and wealth, 1983–2014
- **Step 1 — build the risk measures.** 321 groups = industry of employment × education level. For each group-year, compute the cross-sectional mean, variance, and third moment of the change in log disposable income. Regress each time series on stock returns:

$$\text{Cyclical skewness risk}_g = \frac{\text{Cov}(\mu_{3,g}(\varepsilon), r_s)}{\mathbb{V}(r_s)}$$

- **Step 2 — assign and regress.** Attribute group measures to workers by education × current industry; Tobit of the portfolio equity share on the three risk measures plus controls
- **Identification concern.** Occupation and stock holdings may be jointly determined by risk preference, intelligence, upbringing, expected inheritance
 - **Twin design (Calvet and Sodini, 2014):** repeat within pairs of identical twins

Group-level evidence (Catherine et al., 2024)



Each circle is one of 321 education $\times$ industry groups, sized by group size. Groups facing more cyclical skewness have lower equity shares — through **both** participation (center) and the conditional share (right).

Covariance and countercyclical variance risk show nothing. Only skewness does.

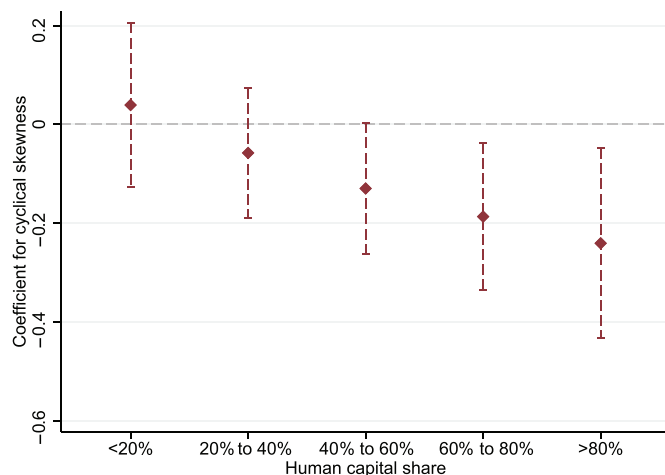
Equity shares within identical twin-year pairs

	(1)	(2)	(3)	(4)	(5)
Δ Cyclical skewness	-0.311 (1.80)			-0.501 (2.66)	-0.422 (2.23)
Δ Count. variance		0.129 (0.46)		0.378 (1.18)	0.504 (1.57)
Δ Covariance			0.108 (0.50)	-0.063 (0.24)	0.081 (0.30)
Δ Skewness				-0.658 (1.20)	-0.628 (1.15)
Δ Variance				-0.024 (0.12)	-0.126 (0.65)
<i>Controls</i>	No	No	No	Yes	Yes
Δ Education FE					Yes
Observations	34,460	34,460	34,460	34,460	34,460
Pseudo R^2	0.000	0.000	0.000	0.044	0.045

Tobit; dependent variable is the **within-twin-year difference** in the equity share. Absolute t -statistics clustered by industry $\times$ education group pairs. Omitted controls: Δ household size, entrepreneur, log assets, and the financial / real estate / debt / pension shares of total wealth.

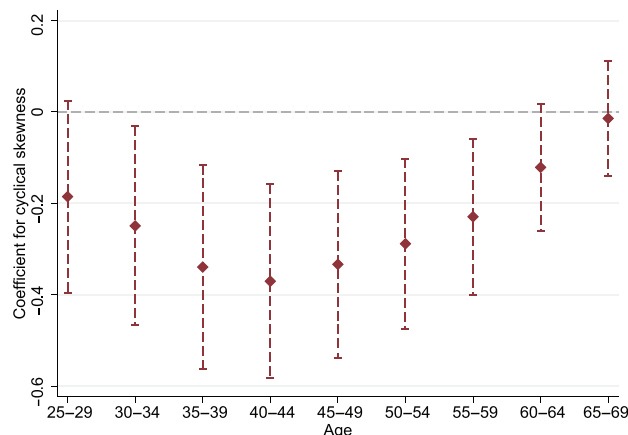
Genes, upbringing and family wealth are differenced out — and the cyclical skewness coefficient survives. Covariance and countercyclical variance still explain nothing.

Prediction 2: the effect scales with the human capital share



- Same Tobit, split by $H/(W + H)$
- The coefficient is **zero** where human capital is a small share of wealth, and grows monotonically in it
- The low- H/W group is a *theoretical placebo*: theory says the effect must vanish there
- An omitted variable explains this only if it happens to be moderated by the human capital share in exactly this way

The model predicted the shape, not just the sign

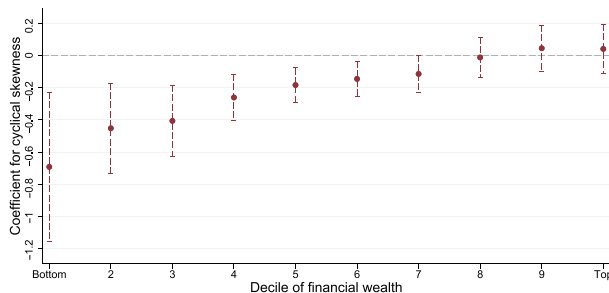


- **U-shaped in age:** near zero in the 20s, trough in the early 40s, back to zero at retirement
- This is exactly Panel B.2 of Figure 7 in [Catherine \(2022\)](#)
- **Why:** with a fixed participation cost, the young hold no stocks *regardless* of skewness; later, the falling H/W ratio takes over
- Magnitude at 40–44: a 2 s.d. difference = 3pp of equity share

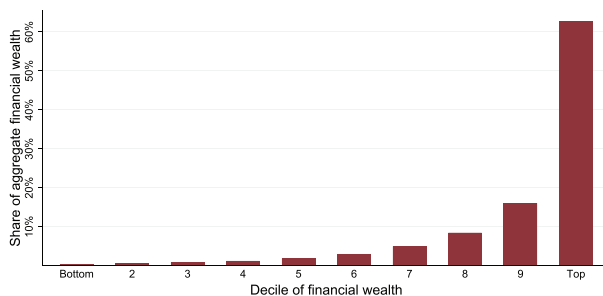
A non-monotonic prediction is a hard test to pass by accident.

Back to asset pricing: two methods, one answer

Panel A. Effect on equity share



Panel B. Share of aggregate financial wealth



- No significant effect in the **top three deciles**; the point estimate is ≈ 25 times smaller than in the first decile
- Those deciles hold **87% of financial wealth** (Panel B) $\Rightarrow$ cyclical skewness in *labor* income cannot move aggregate equity demand much
- Part 1 reached the same conclusion by a completely different route: at most 0.5pp of equity premium
- This *redirects* rather than rejects Schmidt (2016) and Constantinides and Ghosh (2017): look to entrepreneurial risk (Salgado et al., 2019; Heaton and Lucas, 2000)

Takeaways: the economics

- **Human capital is not bond-like — but not because of covariance.**

Its market beta comes from the *cyclical*ity of *tail risk*: the worst career shocks arrive in the years the market falls. Covariance, the moment the empirical literature spent two decades measuring, is ≈ 0 by construction

- **This fixes the life-cycle model** (Catherine, 2022).

With cyclical skewness the model matches wealth, participation and a *rising* equity share at $\gamma \approx 6$, $\beta \approx 0.96$ and a \$250 participation cost. Without it, the same targets need $\gamma = 8$ and \$400 — and still deliver a *falling* profile

- **And households behave as if they know it** (Catherine et al., 2024).

A two-standard-deviation increase in cyclical skewness lowers the equity share by 2–8pp; the effect scales with $H/(W + H)$, survives within identical twin pairs, and vanishes at retirement

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